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Current interrupt devices using shape memory materials

Abstract

Embodiments described herein relate to current interrupt devices (CIDs) for electrochemical cells that use a thermal trigger (e.g., shape memory and/or bi-metallic materials) to open an electrical circuit just prior to a thermal runaway or during short-circuit event to prevent catastrophic failure of the electrochemical cell. Embodiments include CIDs comprising a housing, a bus bar coupled to the housing, and a thermal trigger operably coupled to the bus bar. In some embodiments, the bus bar can include an engineered fracture site. In some embodiments, the thermal trigger is dimensioned and configured to deform at a predetermined temperature to break the bus bar at the engineered fracture site. In some embodiments, a portion of the bus bar travels about a hinge, opening the electrical circuit and preventing overcharging, thermal runaway, and/or other catastrophic failure events.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/048,765 filed on Jul. 30, 2018, titled "Current Interrupt Devices Using Shape Memory Materials," which claims priority to, and the benefit of U.S. Provisional Application No. 62/539,156 filed on Jul. 31, 2017, titled "Current Interrupt Devices Using Shape Memory Materials," the contents of each of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Current interrupt devices (CIDs) are often used as a safety feature for electrochemical cells to

prevent overcharging and/or short-circuiting, which can lead to catastrophic failure of the electrochemical cell. Existing CIDs often use the internal pressure generated within the electrochemical cell as a means to interrupt or disconnect the active cell elements from the electrical current path. This method of current interruption can lead to gas exposure and often can still result in catastrophic failure for the electrochemical cell.

SUMMARY

(2) Embodiments described herein relate to current interrupt devices (CIDs) for electrochemical cells that use a thermal trigger (e.g., shape memory and/or bi-metallic materials) to open an electrical circuit just prior to a thermal runaway or during a short-circuit event to prevent catastrophic failure of the electrochemical cell. Embodiments include CIDs comprising a housing, a bus bar coupled to the housing, and a thermal trigger operably coupled to the bus bar. In some embodiments, the bus bar can include an engineered fracture site. In some embodiments, the thermal trigger is dimensioned and configured to deform at a predetermined temperature to break the bus bar at the engineered fracture site. In some embodiments, a portion of the bus bar travels about a hinge, opening the electrical circuit and preventing overcharging, thermal runaway, and/or other catastrophic failure events.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of a current interrupt device, according to an embodiment.
- (2) FIG. 2 is a perspective view of a current interrupt device, according to an embodiment.
- (3) FIG. 3 is an exploded view of the current interrupt device of FIG. 2.
- (4) FIG. 4A is a cross-sectional view of the current interrupt device of FIG. 2 in a first configuration.
- (5) FIG. 4B is a cross-sectional view of the current interrupt device of FIG. 2 in a second configuration.

DETAILED DESCRIPTION

(6) Embodiments described herein relate generally to safety devices for electrochemical cells, and specifically to a safer and more reliable current interrupt device (CID), the CID configured to discontinue current into or out of the electrochemical cell in response to an increase in the temperature of the electrochemical cell. Specifically, embodiments described herein relate to CIDs for electrochemical cells that use a thermal trigger (e.g., shape memory and/or bi-metallic materials that deform in response to a temperature rising above a predetermined threshold) to open an electrical circuit prior to thermal runaway or during a short-circuit event to prevent catastrophic failure of the electrochemical cell.

(7) In electrochemical cell (e.g., lithium ion battery) design, safety is one of the primary areas of concern. In certain conditions where the cell is overcharged or short circuited, there is a potential for sudden release of stored energy resulting in catastrophic failure (i.e., explosion, fire, flying parts, etc.). To prevent catastrophic failure during these conditions it is desirable to interrupt the flow of energy into and out of the cell prior to thermal runaway or during a short-circuit event, terminating the electrochemical reactions that are causing or accentuating the failure state of the electrochemical cell. This can be accomplished in several ways, including physically disconnecting or breaking the electrical current path into the cell. Typical CIDs used in battery (e.g., lithium battery) applications are very costly and complex to implement, with suspect reliability.

(8) In some embodiments, the CIDs described herein can be configured to disconnect or disrupt the flow of current into the cell once the CIDs or a component thereof (e.g., a thermal trigger) reaches a predetermined temperature threshold. In some embodiments, the CIDs can be engineered to disrupt the flow of current at a variety of temperatures and within a narrow temperature range,

making the “failure” point more precise by reducing the lag between when the temperature reaches the predetermined temperature threshold and when the CID is activated. The CIDs described herein have fewer components than conventional CIDs, thereby increasing reliability and decreasing manufacturing costs. The CIDs described herein can also be disposed inside the electrochemical cell or external to the electrochemical cell and can be easily configured to be used on any existing cell packaging design, battery shape and/or size, and/or discharge/recharge parameters.

(9) In some embodiments, the CIDs described herein can include a conductive element electrically coupled to a terminal of an electrochemical cell. In some embodiments, the CIDs can include a thermal trigger operably coupled to the conductive element. In some embodiments, the thermal trigger can be dimensioned and configured to deform at a predetermined temperature from a first configuration to a second configuration. In some embodiments, the deformation of the thermal trigger from the first configuration to the second configuration can cause the conductive element to fracture. In some embodiments, the conductive element can include an engineered fracture site configured to break as a result of deformation of the thermal trigger from the first configuration to the second configuration. In some embodiments, the conductive element can include a hinge disposed between the fixed portion and the movable portion of the conductive element. In some embodiments, the hinge can define an axis of rotation about which the movable portion of the conductive element travels from the first configuration to the second configuration. In some embodiments, the thermal trigger includes at least one of a shape memory alloy and a bi-metallic material. In some embodiments, the thermal trigger is made from nitinol or a nitinol alloy. In some embodiments, the conductive element is electrically coupled to a terminal of an electrochemical cell. In some embodiments, communication of an electrical current through the terminal is discontinued when the conductive element fractures.

(10) In some embodiments, the CIDs described herein can include a conductive element including a movable portion separated from a fixed portion by a hinge. In some embodiments, the conductive element can be electrically coupled to a terminal of an electrochemical cell. In some embodiments, the CID can further include a thermal trigger positioned a distance from the conductive element in a first configuration. In some embodiments, the thermal trigger can be dimensioned and configured to deform at a predetermined temperature from the first configuration to a second configuration in which the thermal trigger causes the movable portion of the conductive element to move about the hinge. In some embodiments, the thermal trigger can include at least one of a shape memory alloy and a bi-metallic material. In some embodiments, the thermal trigger can be made from nitinol or a nitinol alloy. In some embodiments, communication of an electrical current through the terminal is discontinued when the movable portion of the conductive element moves about the hinge.

(11) FIG. 1 is a schematic illustration of a CID **100** configured to be operably coupled to an electrochemical cell (not shown) to prevent at least one of short-circuiting, thermal runaway, overcharging, an overvoltage condition, or an overcurrent condition. In some embodiments, the CID **100** can be connected to a positive terminal or negative terminal of the electrochemical cell. In some embodiments, the CID **100** or a component thereof is operably integrated into one of the positive terminal or negative terminal of the electrochemical cell. In some embodiments, the CID **100** can be in electrical communication with the electrochemical cell and/or an external electrical source or load. In some embodiments, the CID **100** can be in thermal communication with the electrochemical cell such that the operating temperature of the electrochemical cell, an electrode thereof, or a housing enclosing the electrochemical cell can be substantially similar to the temperature of the CID **100** or a component thereof. In some embodiments, the CID **100** can be configured to allow the transfer of electrical current between the electrochemical cell and a load (e.g., device powered by the electrochemical cell) during discharge or an external source of electrical energy and the electrochemical cell during recharging. In some embodiments, the CID **100** can be maintained in sufficiently close proximity to the electrochemical cell such that thermal energy generated by the electrochemical cell and communicated to the CID **100** will cause the

activation of the CID **100**. In other words, if an electrochemical cell experiences one of, at least one of, or all of a thermal runaway event, a short-circuiting event, an overcharging event, physical damage that causes any of these, or if general overheating due to external temperature change occurs, the CID **100**, being sufficiently positioned nearby the electrochemical cell, can be configured to open the electrical circuit to prevent catastrophic failure of the electrochemical cell. (12) In some embodiments, the CID **100** can include a housing **110**, a bus bar **120** coupled to the housing **110**, and a thermal trigger **130** operably coupled to the bus bar **120**. In some embodiments, the housing **110** can be any suitable shape or form-factor, including but not limited to, square, rectangular, prismatic, cylindrical, any combination thereof, or any other shape or configuration suitable for an electrochemical cell. In some embodiments, the housing **110** can be electrically insulating. In some embodiments, the housing **110** can be configured to be physically coupled to the electrochemical cell. In some embodiments, the housing **110** can have a first aperture (not shown) through which a terminal (not shown) of the bus bar **120** is exposed for connection to a recharging source and/or an outside load. In some embodiments, the housing **110** has a second aperture (not shown) which can allow for a portion of the bus bar **120** to move when the CID **100** is activated.

(13) In some embodiments, the housing **110** can be made from any suitably durable and suitably insulating material. In some embodiments, the housing **110** can be made from any of a polymer, a plastic, ceramic, glass, fiberglass, acrylonitrile-butadiene-styrene, acetate, acrylic, polymerized formaldehyde, epoxy-fiberglass laminate, polystyrene, high impact polystyrene, polyimide, fluoropolymer, polyvinylidene fluoride, melamine laminated with woven glass, unfilled polyimide, mica, rubber, neoprene, aromatic polyamides, nylon, polyetheretherketone, polyethylene terephthalate, glycol-modified polyethylene terephthalate, phenolics, perfluoroalkoxy, polycarbonates, polyesters, polyolefins, polysulfones, polyurethanes, polytetrafluoroethylene, crosslinked polystyrene, polyphenylene sulfide, silicone-fiberglass resin, unfilled polyetherimide, vulcanized rubber, vulcanized fiber, dacron, mylar, polyvinylfluoride, polychlorinated biphenyls, and combinations thereof.

(14) In some embodiments, the bus bar **120** can be disposed on or substantially on the housing **110**. In some embodiments, the bus bar **120** can be disposed within, substantially within, at least partially within the housing **110**. In some embodiments, the bus bar **120** can be dimensioned and configured to allow the transfer of electrical energy between the electrochemical cell and the external source during recharging of the electrochemical cell or between the electrochemical cell and the external load (e.g., device being powered by the electrochemical cell) during discharge. In some embodiments, the bus bar **120** can be operably coupled to the positive terminal of the electrochemical cell or the negative terminal of the electrochemical cell. In some embodiments, the bus bar **120** can connect one of the positive terminal and the negative terminal of the electrochemical cell to a system level bus (e.g., of a battery pack or battery module).

(15) In some embodiments, the bus bar **120** can be made from any conductive material that is durable enough to remain intact during normal operation of the electrochemical cell and that will fracture at an engineered fracture site (not shown) due to mechanical stress. In some embodiments, the bus bar **120** can be made from any of silver, copper, gold, aluminum, molybdenum, zinc, lithium, tungsten, brass, carbon-based materials, nickel, iron, palladium, platinum, tin, bronze, carbon steel, lead, titanium, stainless steel, any other suitable conductive material, and any alloy, admixture, or combination thereof.

(16) In some embodiments, the bus bar **120** can include a fixed portion (not shown) and a movable portion (not shown). The fixed portion can be coupled to the housing **110**, and the movable portion can be substantially unconstrained by the housing **110**. In other words, the fixed portion can be configured to be permanently coupled to the housing **110**, and the movable portion can be moved with respect to the housing **110** and/or the fixed portion in response to an external force (e.g., from the thermal trigger **130**). In some embodiments, the displacement of the movable portion from a

first position to a second position may cause the bus bar **120** to break, thereby separating the movable portion of the bus bar **120** from the fixed portion of the bus bar **120** and disconnecting the electrical circuit, thereby preventing any further transfer of electrical energy into or out of the electrochemical cell. In some embodiments, the bus bar **120** can be configured to break at the engineered fracture site. In some embodiments, the engineered fracture site can be a groove, etched line, furrow, rut, trench, indentation, hollow, or other such depression. In some embodiments, the engineered fracture site can be a thinned portion of the bus bar **120**. In some embodiments, the bus bar **120** can be tempered or otherwise treated to improve mechanical properties, with the exception of the engineered fracture site such that bus bar **120** is designed to break at the engineered fracture site before mechanical failure of other portions of the bus bar **120** occur. In some embodiments, the bus bar **120** can be formed from a first material or materials and the engineered fracture site can be formed from a second material or materials, the second material or materials having a mechanical property different than the first material or materials such that the bus bar **120** fails at the engineered fracture site under mechanical stress. In some embodiments, once the bus bar **120** is broken (e.g., at the engineered fracture site) and the electrical current into or out of the electrochemical cell is discontinued, the detrimental thermal and/or electrical event that was occurring is stopped, substantially stopped, slowed, or temporarily discontinued and catastrophic failure is avoided.

(17) In some embodiments, the thermal trigger **130** can be coupled to the housing **110** and/or the bus bar **120** and configured to deform at a predetermined temperature threshold or within a predetermined temperature threshold range. In some embodiments, the thermal trigger **130** can be positioned and configured such that deformation of the thermal trigger **130** supplies mechanical stress to the movable portion of the bus bar **120**. In some embodiments, the mechanical stress supplied to the movable portion of the bus bar **120** can cause the bus bar **120** to break or substantially break at the engineered fracture site or elsewhere. In some embodiments, the mechanical stress supplied to the movable portion of the bus bar **120** can cause the displacement of the movable portion of the bus bar **120**, causing the movable portion to move away from the fixed portion and discontinuing electrical current into or out of the electrochemical cell. In some embodiments, the thermal trigger **130** causes the bus bar **120** to fracture due to mechanical stress caused by controlled deformation of the thermal trigger **130** in response to temperature change. In some embodiments, the fracture can be any of a brittle fracture, a ductile fracture, a slip fracture, a creep rupture failure, buckling failure, any other failure type, or combinations thereof. In some embodiments, the thermal trigger **130** can deform at the predetermined temperature, causing an intermediary feature, mechanism, or device to cause the bus bar **120** to fracture. In some embodiments, the thermal trigger **130** can deform at the predetermined temperature, causing the movable portion of the bus bar **120** to be moved translationally or rectilinearly away from the terminal. In some embodiments, deformation of the thermal trigger **130** at the predetermined temperature can initiate a chemical and/or thermal event that results in the disconnection of the electrical circuit. In some embodiments, deformation of the thermal trigger **130** at the predetermined temperature can cause the bus bar **120** to short-circuit, discontinuing electrical current coming from or going to the electrochemical cell. In some embodiments, the bus bar **120** can be engineered to have a precise strain resistance and to be brittle such that deformation of the thermal trigger **130** at the predetermined temperature can cause the bus bar **120** to shatter. In some embodiments, the bus bar **120** can be engineered to be at least somewhat resistant to electrical current such that, above a particular current level and/or voltage, the bus bar **120** can heat up, causing the thermal trigger **130** to deform and break the bus bar **120**.

(18) In some embodiments, the thermal trigger **130** can be engineered to transition between a plurality of configurations in response to the temperature of the thermal trigger **130** reaching or exceeding a predetermined temperature or a predetermined temperature range. In some embodiments, the thermal trigger **130** can have a first configuration below the predetermined

temperature or predetermined temperature range, and second configuration at or above the predetermined temperature or predetermined temperature range. In some embodiments, the first configuration is a bent configuration. In some embodiments, the second configuration is an extended configuration. In some embodiments, the first configuration is an extended configuration and the second configuration is a bent configuration. In some embodiments, the first configuration is a more tightly coiled configuration and the second configuration is a less tightly coiled configuration. In some embodiments, the first configuration is a compressed configuration and the second configuration is a stretched or extended configuration. In some embodiments, the first configuration is a folded configuration and the second configuration is an unfolded configuration. In some embodiments, the first configuration is a wound configuration and the second configuration is an unwound configuration. In some embodiments, the first configuration is a flat configuration and the second configuration is a rounded configuration. In some embodiments, the first configuration is an unstructured configuration and the second configuration is a structured configuration. In some embodiments, the first configuration is a compressible configuration and the second configuration is a non-compressible configuration. In some embodiments, the first configuration is a deflated configuration and the second configuration is an inflated configuration. In some embodiments, the first configuration is a rounded configuration and the second configuration is a squared-off configuration. In some embodiments, the thermal trigger **130** can be superelastic at temperatures below the predetermined temperature and revert back to a straight annealed shape at or above the predetermined temperature.

(19) In some embodiments, the thermal trigger **130** can include or be formed from a shape memory material such that the thermal trigger **130** has shape memory characteristics. In some embodiments, the shape memory characteristics of the thermal trigger **130** can lead to irreversible deformation. In other words, in some embodiments, the thermal trigger **130** can move from the first configuration to the second configuration at the predetermined temperature, however if the temperature subsequently drops below the predetermined temperature the thermal trigger **130** may remain in the second configuration. In some embodiments, the thermal trigger **130** has reversible shape memory characteristics, meaning the thermal trigger **130** may remain in the second configuration while the temperature of the thermal trigger **130** remains at or above the predetermined temperature and may return to the first configuration once the temperature returns to below the predetermined temperature.

(20) In some embodiments, deformation of the thermal trigger **130** causes the movable portion of the bus bar **120** to move about an axis in a hinge-like fashion. In some embodiments, the axis is a hinge (not shown) in the bus bar **120**. In some embodiments, the hinge can be approximately parallel to the engineered fracture site. In some embodiments, the distance of travel of the movable portion of the bus bar **120** from the fixed portion of the bus bar **120** is determined at least in part by the distance of travel of the thermal trigger **130**. In some embodiments, the distance of travel of the movable portion of the bus bar **120** can also be determined in part due to the distance between the hinge and the engineered fracture site. In some embodiments, the hinge in the bus bar **120** can be defined by a weakening of a portion of the bus bar **120**, for example by removing material from the bus bar **120**. In some embodiments, the hinge can be defined by scoring the portion of the bus bar **120**. In some embodiments, the hinge can be defined by heat treating the portion of the bus bar **120**. In some embodiments, the hinge can be defined by chemically treating the portion of the bus bar **120**. In some embodiments, the bus bar **120** can be formed from a first portion and a second portion, the first and second portions coupled at a region, the coupling region defining a natural hinge point. In some embodiments, the portion of the bus bar **120** can be mechanically worked such that stress/strain impacts can cause the formation of the hinge.

(21) In some embodiments, the thermal trigger **130** can be positioned between the housing **110** and the bus bar **120** such that the extent and direction of travel of the thermal trigger **130** due to thermal deformation of the thermal trigger **130** is controlled by the position of the thermal trigger **130**

relative to the housing **110** and the bus bar **120**. In some embodiments, the thermal trigger **130** can be coupled to the housing **110**. In some embodiments, the thermal trigger **130** can be coupled to the bus bar **120**. In some embodiments, the thermal trigger **130** can be coupled to the housing **110** and the bus bar **120**. In some embodiments, the housing **110** and/or the bus bar **120** can include a cavity (not shown) into which the thermal trigger **130** can be at least partially disposed. In some embodiments, the thermal trigger **130** is adhered to the housing **110** and/or the bus bar **120** using a chemical bonding agent, a thermal fusing process, a high frictional coefficient caused by low engineering tolerance between the size of the cavity and the size of the thermal trigger **130**, and any other suitable process or combination thereof. In some embodiments, the housing **110** can include a notch (not shown) that corresponds to the hinge on the bus bar **120** such that the notch and the hinge together provide an axis about which the movable portion of the bus bar **120** rotates after fracture of the bus bar **120**.

(22) In some embodiments, the thermal trigger **130** is made from a material that deforms predictably at a particular temperature. In some embodiments, the material used to form the thermal trigger **130** can be shape memory alloys (SMAs), nickel titanium alloys, nitinol alloys, nitinol 55, nitinol 60, nitinol-copper alloys, nitinol-niobium alloys, copper-aluminum-nickel alloys, zinc-copper-gold-iron alloys, Fe—Mn—Si alloys, Cu—Zn—Al alloys, Cu—Al—Ni alloys, Ag—Cd, Ag—Cd 44/49 at. % Cd, Au—Cd, Au—Cd 46.5/50 at. % Cd, Cu—Al—Ni, Cu—Al—Ni 14/14.5 wt % Al and 3/4.5 wt % Ni, Cu—Sn, Cu—Sn having approximately 15 at % Sn, Cu—Zn, Cu—Zn 38.5/41.5 wt. % Zn, Cu—Zn—X (where X is at least one of Si, Al, or Sn), Fe—Pt, Fe—Pt having approximately 25 at. % Pt, Mn—Cu, Mn—Cu 5/35 at % Cu, Fe—Mn—Si, Co—Ni—Al, Co—Ni—Ga, Ni—Fe—Ga, Ti—Nb, Ni—Ti, Ni—Ti having approximately 55-60 wt % Ni, Ni—Ti—Hf, Ni—Ti—Pd, Ni—Mn—Ga, any suitable bi-metallic alloy, Cu—Cu, Al—Al, and any combination thereof.

(23) In some embodiments, the thermal trigger **130** can be any suitable shape such that configurational change from the first configuration to the second configuration achieves the desired fracture of the bus bar **120** and desired distance of travel of the movable portion. In some embodiments, suitable shapes for the thermal trigger **130** can include any of rectangular, cylindrical, square, tab, oblong, circular, triangular, wire, sheet, rod, ribbon, tube, ring, foil, helical coil springs, or any combination thereof. In some embodiments, the thermal trigger **130** can be formed using any suitable method, including but not limited to hot working, cold working, super elastic strain annealing, flat strain annealing, shape set strain annealing, or any combination thereof. In some embodiments, the thermal trigger **130** may form a natural oxide on the surface, for instance TiO_2 in the case of nitinol and nitinol alloys.

(24) In some embodiments, the thermal trigger **130** can be a single element. In some embodiments, the thermal trigger **130** can be a plurality of elements. In some embodiments, the plurality of elements that comprise the thermal trigger **130** can be two, three, four, five, six or more elements. In some embodiments, the thermal trigger **130** can be a single component having a single portion. In some embodiments, the thermal trigger **130** can include a plurality of components operably coupled to form the thermal trigger **130**. In some embodiments, the thermal trigger **130** can include a plurality of portions, for example of the same material, coupled to form the thermal trigger **130**. In some embodiments, the thermal trigger **130** can be coated in a material to make it electrically insulating. For example, the thermal trigger **130** can be coated in a polymer, a synthetic rubber material, a natural rubber material, a ceramic coating, an organic material, an organometallic, a non-conductive metal, or a metal that is less conductive than the thermal trigger **130** itself. In some embodiments, the thermal trigger **130** can include or be formed from a material or a plurality of materials that is or are electrically insulating.

(25) In some embodiments, the predetermined temperature at which the thermal trigger **130** deforms can be between about 40° C. and about 200° C., between about 45° C. and about 190° C., between about 50° C. and about 180° C., between about 55° C. and about 170° C., between about

60° C. and about 160° C., between about 65° C. and about 150° C., between about 70° C. and about 145° C., between about 75° C. and about 140° C., between about 80° C. and about 135° C., between about 85° C. and about 130° C., between about 90° C. and about 125° C., between about 95° C. and about 120° C., between about 100° C. and about 115° C., between about 40° C. and about 190° C., between about 40° C. and about 180° C., between about 40° C. and about 170° C., between about 40° C. and about 160° C., between about 40° C. and about 150° C., between about 40° C. and about 140° C., between about 40° C. and about 130° C., between about 40° C. and about 120° C., between about 40° C. and about 110° C., between about 40° C. and about 100° C., between about 40° C. and about 90° C., between about 40° C. and about 80° C., between about 40° C. and about 70° C., between about 40° C. and about 60° C., between about 40° C. and about 50° C., between about 50° C. and about 200° C., between about 60° C. and about 200° C., between about 70° C. and about 200° C., between about 80° C. and about 200° C., between about 90° C. and about 200° C., between about 100° C. and about 200° C., between about 110° C. and about 200° C., between about 120° C. and about 200° C., between about 130° C. and about 200° C., between about 140° C. and about 200° C., between about 150° C. and about 200° C., between about 160° C. and about 200° C., between about 170° C. and about 200° C., between about 180° C. and about 200° C., or between about 190° C. and about 200° C., inclusive of all values and ranges therebetween. In some embodiments, the predetermined temperature at which the thermal trigger **130** deforms can be above about 45° C., above about 50° C., above about 55° C., above about 60° C., above about 65° C., above about 70° C., above about 75° C., above about 80° C., above about 85° C., above about 90° C., above about 95° C., above about 100° C., above about 105° C., above about 110° C., above about 115° C., above about 120° C., above about 125° C., above about 130° C., above about 135° C., above about 140° C., above about 145° C., and above about 150° C., inclusive of all ranges and values therebetween. In some embodiments, the predetermined temperature at which the thermal trigger **130** deforms can be below about 200° C., below about 190° C., below about 180° C., below about 170° C., below about 160° C., below about 150° C., below about 145° C., below about 140° C., below about 135° C., below about 130° C., below about 125° C., below about 120° C., below about 115° C., below about 110° C., below about 105° C., below about 100° C., below about 95° C., below about 90° C., below about 85° C., below about 80° C., below about 75° C., below about 70° C., below about 65° C., below about 60° C., below about 55° C., below about 50° C., below about 45° C., or below about 40° C., inclusive of all values and ranges therebetween.

(26) In some embodiments, the thermal trigger **130** predictably deforms within a range of temperatures around the predetermined temperature. In some embodiments, the range of temperatures is $\pm 1^\circ$ C., $\pm 2^\circ$ C., $\pm 3^\circ$ C., $\pm 4^\circ$ C., 5° C., $\pm 6^\circ$ C., $\pm 7^\circ$ C., $\pm 8^\circ$ C., $\pm 9^\circ$ C., or $\pm 10^\circ$ C., inclusive of all ranges and values therebetween.

(27) In some embodiments, the distance of travel of the movable portion of the bus bar **120** can be caused either directly or indirectly by the deformation of the thermal trigger **130**. In some embodiments, the distance of travel of the movable portion of the bus bar **120** can be less than about 0.1 mm, 0.2 mm, 0.3 mm, 0.4 mm, 0.5 mm, 0.6 mm, 0.7 mm, 0.8 mm, 0.9 mm, 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, 11 mm, 12 mm, 13 mm, 14 mm, or 15 mm, inclusive of all values and ranges therebetween. In some embodiments, the distance of travel of the movable portion of the bus bar **120** can be greater than about 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, 11 mm, 12 mm, 13 mm, 14 mm, or 15 mm, inclusive of all values and ranges therebetween. In some embodiments, the distance of travel of the movable portion of the bus bar **120** can be between about 0.1 mm and about 15 mm, about 0.2 mm and about 14 mm, about 0.3 mm and about 13 mm, about 0.4 mm and about 12 mm, about 0.5 mm and about 11 mm, about 1 mm and about 10 mm, about 2 mm and about 9 mm, about 3 mm and about 8 mm, about 4 mm and about 7 mm, about 5 mm and about 6 mm, about 0.1 mm and about 14 mm, about 0.1 mm and about 13 mm, about 0.1 mm and about 12 mm, about 0.1 mm and about 11 mm,

about 0.1 mm and about 10 mm, about 0.1 mm and about 9 mm, about 0.1 mm and about 8 mm, about 0.1 mm and about 7 mm, about 0.1 mm and about 6 mm, about 0.1 mm and about 5 mm, about 0.1 mm and about 4 mm, about 0.1 mm and about 3 mm, about 0.1 mm and about 2 mm, about 0.1 mm and about 1 mm, about 0.1 mm and about 0.9 mm, about 0.1 mm and about 0.8 mm, about 0.1 mm and about 0.7 mm, about 0.1 mm and about 0.6 mm, about 0.1 mm and about 0.5 mm, about 0.1 mm and about 0.4 mm, about 0.1 mm and about 0.3 mm, about 0.1 mm and about 0.2 mm, about 0.2 mm and about 15 mm, about 0.3 mm and about 15 mm, about 0.4 mm and about 15 mm, about 0.5 mm and about 15 mm, about 0.6 mm and about 15 mm, about 0.7 mm and about 15 mm, about 0.8 mm and about 15 mm, about 0.9 mm and about 15 mm, about 1 mm and about 15 mm, about 2 mm and about 15 mm, about 3 mm and about 15 mm, about 4 mm and about 15 mm, about 5 mm and about 15 mm, about 6 mm and about 15 mm, about 7 mm and about 15 mm, about 8 mm and about 15 mm, about 9 mm and about 15 mm, about 10 mm and about 15 mm, about 11 mm and about 15 mm, about 12 mm and about 15 mm, about 13 mm and about 15 mm, or about 14 mm and about 15 mm, inclusive of all values and ranges therebetween.

(28) In some embodiments, the duration of time within which the thermal trigger **130** moves from the first configuration to the second configuration can be less than about 1 second, less than about 2 seconds, less than about 3 seconds, less than about 4 seconds, less than about 5 seconds, less than about 6 seconds, less than about 7 seconds, less than about 8 seconds, less than about 9 seconds, less than about 10 seconds, less than about 15 seconds, less than about 20 seconds, less than about 25 seconds, less than about 30 seconds, or less than about 1 minute, inclusive of all ranges and values therebetween. In some embodiments, the duration of time within which the thermal trigger **130** moves from the first configuration to the second configuration can be greater than about 1 seconds, greater than about 2 seconds, greater than about 3 seconds, greater than about 4 seconds, greater than about 5 seconds, greater than about 6 seconds, greater than about 7 seconds, greater than about 8 seconds, greater than about 9 seconds, greater than about 10 seconds, greater than about 15 seconds, greater than about 20 seconds, greater than about 25 seconds, greater than about 30 seconds, or greater than about 1 minute, inclusive of all values and ranges therebetween.

(29) In some embodiments, there can be a time lag between when the thermal trigger **130** reaches or exceeds the predetermined temperature or predetermined temperature range and when the thermal trigger **130** transitions from the first configuration to the second configuration. In some embodiments, the time lag can be less than about 30 seconds, less than about 25 seconds, less than about 20 seconds, less than about 15 seconds, less than about 10 seconds, less than about 9 seconds, less than about 8 seconds, less than about 7 seconds, less than about 6 seconds, less than about 5 seconds, less than about 4 seconds, less than about 3 seconds, less than about 2 seconds, or less than about 1 second, inclusive of all values and ranges therebetween.

(30) By way of example only, nitinol can be used as the thermal trigger **130**, according to an embodiment. Nitinol is a superelastic shape memory alloy at standard temperatures and therefore can be easily manipulated into any shape or configuration in the Martensitic form. However, when the alloy is heated to above a set temperature the nitinol reverts back to the Austenitic shape. Whereas the Martensitic form of nitinol is weak and undergoes strain in response to stress, the Austenitic shape is very strong, having a yield strength of between about 35,000 and 100,000 psi depending upon the alloy composition. Austenitic nitinol may have a Young's modulus of about 83 GPa, an ultimate tensile strength after being work hardened of 1900 MPa, a Poisson's ratio of about 0.33, and an elongation at failure of between about 5% and about 10%. Nitinol wire may be heat treated in the extended, second configuration while constrained to that shape and then cooled to room temperature such that it can easily be formed into the desired first configuration at room temperature and disposed within the housing **110** of the CID **100**. Heat treating of nitinol may be accomplished at temperatures as low as 400° C. and times as short as 1-2 minutes, however heat treating can also require a temperature closer to 500° C. and times over 5 minutes. The heat treated nitinol can then be rapidly cooled via a water quench in order to set the Austenitic shape. Adjusting

the heat treating temperature and duration may change the temperature at which the nitinol transitions from the Martensitic to Austenitic shape. Changing the concentration of nickel to titanium in the nitinol, even slightly, can also result in very large changes in the temperature at which nitinol changes from the Martensitic to Austenitic shape. Once heat treated and subsequently cooled, the nitinol is again malleable and can be shaped in to the desired first configuration. Upon an increase in the temperature of the nitinol wire within the CID **100** due to a thermal event within an electrochemical cell the nitinol wire reverts back to the second configuration quickly enough and with a high enough yield strength to fracture the bus bar **120** at the engineered fracture site and open the electrical circuit.

(31) In some embodiments, the CID **100** is operably coupled to an electrochemical cell such that the CID **100** can be activated to discontinue discharge or charging of the electrochemical cell in response to over-temperature conditions. In some embodiments, the actuation of the thermal trigger **130** leading to the opening of the electrical circuit may occur after an electrochemical cell to which the CID **100** is operably coupled has already been irreversibly damaged. In some embodiments, the actuation of the thermal trigger **130** occurs at a low enough temperature such that an electrochemical cell to which the CID **100** is operably coupled is not irreversibly damaged. In other words, in some embodiments, activation of the CID **100** may reduce the risk of irreversible damage to the electrochemical cell. In some embodiments, the CID **100** can be resettable or replaceable. In some embodiments, if the CID **100** is activated and electrochemical cell is protected from irreversible damage or catastrophic failure (e.g., thermal runaway), the CID **100** or a component thereof can be replaced and the electrochemical cell can be returned to operation. In some embodiments, the whole CID **100** can be replaced to return the electrochemical cell to operation. In some embodiments, the bus bar **120** can be replaced to return the CID **100** to its original pre-activation condition and to return to the electrochemical cell to operation. In some embodiments, the bus bar **120** and the thermal trigger **130** can be replaced to return the CID **100** to its original pre-activation condition and to return the electrochemical cell to operation.

(32) FIGS. **2**, **3**, and **4A-4B** illustrate a CID **200** similar to the CID **100** described above, being configured to improve safety and reduce the likelihood of overcharging and/or short-circuiting in an electrochemical cell. In some embodiments, portions and/or aspects of the CID **200** are substantially similar in form and/or function to the corresponding portions and/or aspects of the CID **100** described above with reference to FIG. **1**. Accordingly, such similar portions and/or aspects are not described in further detail herein.

(33) In some embodiments, the CID **200** is connected to a positive or negative lead of an electrochemical cell. In some embodiments, the CID **200** or a component thereof is the positive or negative lead of an electrochemical cell. In other words, the CID **200** can be in electrical communication with an electrochemical cell and/or an external electrical source or load. The CID **200** is configured to allow the transfer electrical energy between an electrochemical cell and a device being powered or an external source of electrical energy and the electrochemical cell during recharging. In some embodiments, the CID **200** can be maintained in sufficiently close proximity to an electrochemical cell such that any thermal energy discharge from the electrochemical cell will cause the CID **200** to undergo a timely temperature change. In other words, if an electrochemical cell experiences any of a thermal runaway event, a short-circuiting event, an overcharging event, over-current conditions, over-voltage conditions, physical damage that causes any of these, or if general overheating due to external temperature change occurs, the CID **200** can be positioned nearby the electrochemical cell and can open the electrical circuit to prevent catastrophic failure of the electrochemical cell. In some embodiments, the CID **200** can include a resistor (not shown) such that during over-current or over-voltage conditions, the resistor can heat up, causing the thermal trigger **230** to heat up to at or above the predetermined temperature. In some embodiments, a component of the CID **200**, e.g., the bus bar **230**, can have resistive properties or characteristics such that above a predetermined current level or a predetermined voltage level, the resistive

component can heat up, causing the thermal trigger **230** to heat up to at or above the predetermined temperature.

(34) In some embodiments, the CID **200** includes a housing **210**, a bus bar **220** coupled to the housing **210**, and a thermal trigger **230** operably coupled to the bus bar **220**. In some embodiments, the housing **210** can be any suitable shape or form-factor, including but not limited to, square, rectangular, prismatic, cylindrical, any combination thereof, or any other shape or configuration suitable for an electrochemical cell. In some embodiments, the housing **210** can be electrically insulating. In some embodiments, the housing **210** can be configured to easily attach to an electrochemical cell.

(35) In some embodiments, the bus bar **220** is disposed on or at least partially in the housing **210**, and is configured to allow the transfer of electrical energy between an electrochemical cell and the external source during recharge or external load during discharge. In some embodiments, the bus bar **220** is dimensioned and configured to connect to either the positive or negative lead of an electrochemical cell. In some embodiments, the bus bar **220** can connect a lead of an electrochemical cell and a system level busing (e.g., of a battery pack or battery module).

(36) In some embodiments, the bus bar **220** can include a terminal **221**, a movable portion **226**, and a fixed portion **228**. The fixed portion **228** can be coupled to the housing **210**, and the movable portion **226** can be substantially unconstrained by the housing **210**. In other words, the fixed portion **228** can be configured to be permanently coupled to the housing **210**, and the movable portion **226** can be moved with respect to the housing **210** and/or the fixed portion **228** in response to an external force (e.g., from the thermal trigger **230**).

(37) In some embodiments, the housing **210** is dimensioned and configured to have a guide **212** into which a bus bar **220** can be slidably disposed. In some embodiments, the housing **210** has a first aperture **214** through which the terminal **221** of the bus bar **220** is exposed for connection to a recharging source and/or outside load. In some embodiments, the housing **210** has a second aperture **216** configured to be aligned with the movable portion **226** of the bus bar **220** to allow the movable portion **226** to be moved through the second aperture **216**. The displacement of the movable portion **226** from a first position to a second position causes the bus bar **220** to break, thereby opening the electrical circuit and preventing any further transfer of electrical energy into or out of the electrochemical cell. In some embodiments, the bus bar **220** can be configured to break at an engineered fracture site **222**. In some embodiments, the engineered fracture site **222** can be a groove, furrow, rut, trench, indentation, hollow, or other such depression. Once the bus bar **220** is broken and the electrical circuit is open, the detrimental thermal and/or electrical event that was occurring is stopped and catastrophic failure of the electrochemical cell is avoided.

(38) The thermal trigger **230** can be coupled to the housing **210** and/or the bus bar **220** and is configured to deform at a predetermined temperature or within a predetermined temperature range. Deformation of the thermal trigger **230** can cause displacement of the movable portion **226** of the bus bar **220**, thereby causing the electrical circuit formed in part by the bus bar **220** to open. In some embodiments, the thermal trigger **230** causes the bus bar **220** to fracture due to mechanical stress caused by controlled deformation of the thermal trigger **230** in response to temperature change. In some embodiments, the fracture can be any of a brittle fracture, a ductile fracture, a slip fracture, a creep rupture failure, buckling failure, any other failure type, or combinations thereof. In some embodiments, the thermal trigger **230** can deform at the predetermined temperature, causing an intermediary feature, mechanism, or device to cause the bus bar **220** to fracture. In some embodiments, the thermal trigger **230** can deform at the predetermined temperature, causing the movable portion **226** of the bus bar **220** to be moved translationally or rectilinearly away from the terminal **221**. In some embodiments, deformation of the thermal trigger **230** at the predetermined temperature can initiate a chemical and/or thermal event that results in the disconnection of the electrical circuit between the electrochemical cell and a charging source or a discharge load.

(39) In some embodiments, the thermal trigger **230** can have a first configuration below a

predetermined temperature or predetermined temperature range, and a second configuration at or above the predetermined temperature or predetermined temperature range. In some embodiments, the first configuration is a bent configuration, and the second configuration is an extended configuration. In some embodiments, the thermal trigger **230** can be superelastic at temperatures below the predetermined temperature and revert back to a straight annealed shape at or above the predetermined temperature. In some embodiments, the shape memory characteristic of the thermal trigger **230** leads to irreversible deformation. In other words, in some embodiments, the thermal trigger **230** can move from the first configuration to the second configuration at the predetermined temperature, however if the temperature subsequently drops below the predetermined temperature the thermal trigger **230** may remain in the second configuration. In some embodiments, the thermal trigger **230** has reversible shape memory, meaning the thermal trigger **230** may only remain in the second configuration while the temperature of the thermal trigger remains at or above the predetermined temperature. In some embodiments, the actuation of the thermal trigger **230** leading to the opening of the electrical circuit occurs after an electrochemical cell has already been irreversibly damaged. In some embodiments, the actuation of the thermal trigger **230** occurs at a low enough temperature such that an electrochemical cell is not irreversibly damaged.

(40) In some embodiments, deformation of the thermal trigger **230** causes the movable portion **226** of the bus bar **220** to move about an axis in a hinge-like fashion. In some embodiments, the axis is a hinge **224** in the bus bar **220**. In some embodiments, the hinge **224** can be positioned approximately parallel to the engineered fracture site **226**. In some embodiments, the distance of travel of the movable portion **226** of the bus bar **220** from the fixed portion **228** of the bus bar **220** is determined at least in part by the distance of travel of the thermal trigger **230**. In some embodiments, the distance of travel of the movable portion **226** of the bus bar **220** can also be determined in part due to the distance between the hinge **224** and the engineered fracture site **222**.

(41) In some embodiments, the thermal trigger **230** can be positioned between the housing **210** and the bus bar **220** such that the extent and direction of travel of the thermal trigger **230** due to thermal deformation of the thermal trigger **230** is controlled by the position of the thermal trigger **230** relative to the housing **210** and the bus bar **220**. In some embodiments, the thermal trigger **230** can be coupled to the housing **210**. In some embodiments, the thermal trigger **230** can be coupled to the bus bar **220**. In some embodiments, the thermal trigger **230** can be coupled to the housing **210** and the bus bar **220**. In some embodiments, the housing **210** and/or the bus bar **220** can include a cavity **218** into which the thermal trigger **230** can be at least partially disposed. In some embodiments, the cavity **218** is dimensioned and configured such that the thermal trigger **230** is fully disposed in the cavity **218** when the thermal trigger **230** is in the first configuration. In some embodiments, when the thermal trigger **230** moves from the first configuration to the second configuration, a portion of the thermal trigger **230** remains disposed in the cavity **218** while a portion of the thermal trigger **230** moves to displace the movable portion **226** of the bus bar **220**. In some embodiments, the thermal trigger **230** is adhered to the housing **210** and/or the bus bar **220** using a chemical bonding agent, a thermal fusing process, a high frictional coefficient caused by low engineering tolerance between the size of the cavity **218** and the size of the thermal trigger **230**, and any other suitable process or combination thereof. In some embodiments, the housing **210** can include a notch (not shown) that corresponds to the hinge **224** of the bus bar **220** such that the notch and the hinge **224** together provide an axis about which the movable portion **226** of the bus bar **220** rotates after fracture of the bus bar **220**.

(42) In some embodiments, the CID **200** is operably coupled to the electrochemical cell such that the CID **200** can be activated to discontinue discharge or charging of the electrochemical cell in response to over-temperature conditions. In some embodiments, the actuation of the thermal trigger **230** leading to the opening of the electrical circuit may occur after an electrochemical cell to which the CID **200** is operably coupled has already been irreversibly damaged. In some embodiments, the actuation of the thermal trigger **230** occurs at a low enough temperature such that an

electrochemical cell to which the CID **200** is operably coupled is not irreversibly damaged. In other words, in some embodiments, activation of the CID **200** may reduce the risk of irreversible damage to the electrochemical cell. In some embodiments, the CID **100** can be resettable or replaceable. In some embodiments, if the CID **200** is activated and electrochemical cell is protected from irreversible damage or catastrophic failure (e.g., thermal runaway), the CID **200** or a component thereof can be replaced and the electrochemical cell can be returned to operation. In some embodiments, the whole CID **200** can be replaced to return the electrochemical cell to operation. In some embodiments, the bus bar **220** can be replaced to return the CID **200** to its original pre-activation condition and to return to the electrochemical cell to operation. In some embodiments, the bus bar **220** and the thermal trigger **230** can be replaced to return the CID **200** to its original pre-activation condition and to return the electrochemical cell to operation.

(43) To provide an overall understanding, certain illustrative embodiments have been described; however, it will be understood by one of ordinary skill in the art that the systems, apparatuses, and methods described herein can be adapted and modified to provide systems, apparatuses, and methods for other suitable applications and that other additions and modifications can be made without departing from the scope of the systems, apparatuses, and methods described herein.

(44) The embodiments described herein have been particularly shown and described, but it will be understood that various changes in form and details may be made. Unless otherwise specified, the illustrated embodiments can be understood as providing exemplary features of varying detail of certain embodiments, and therefore, unless otherwise specified, features, components, modules, and/or aspects of the illustrations can be otherwise combined, separated, interchanged, and/or rearranged without departing from the disclosed systems or methods. Additionally, the shapes and sizes of components are also exemplary and unless otherwise specified, can be altered without affecting the scope of the disclosed and exemplary systems, apparatuses, or methods of the present disclosure.

(45) As used herein, the term “about” and “approximately” generally mean plus or minus 10% of the value stated, for example about 250 μm would include 225 μm to 275 μm , approximately 1,000 μm would include 900 μm to 1,100 μm .

(46) Conventional terms in the fields of endoscopic devices have been used herein. The terms are known in the art and are provided only as a non-limiting example for convenience purposes. Accordingly, the interpretation of the corresponding terms in the claims, unless stated otherwise, is not limited to any particular definition. Thus, the terms used in the claims should be given their broadest reasonable interpretation.

(47) Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that any arrangement that is adapted to achieve the same purpose may be substituted for the specific embodiments shown. Many adaptations will be apparent to those of ordinary skill in the art. Accordingly, this application is intended to cover any adaptations or variations.

(48) The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments that may be practiced. These embodiments are also referred to herein as “examples.” Such examples may include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

(49) All publications, patents, and patent documents referred to in this document are incorporated by reference herein in their entirety, as though individually incorporated by reference. In the event of inconsistent usages between this document and those documents so incorporated by reference,

the usage in the incorporated reference(s) should be considered supplementary to that of this document; for irreconcilable inconsistencies, the usage in this document controls.

(50) In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

(51) The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments may be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure and is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims.

(52) In this Detailed Description, various features may have been grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments may be combined with each other in various combinations or permutations. The scope of the embodiments should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A current-interrupt device (CID), comprising: a conductive element configured to be electrically coupled to a terminal of an electrochemical cell, the conductive element including a thinned portion; and a thermal trigger operably coupled to the conductive element, the thermal trigger dimensioned and configured to deform at a predetermined temperature from a first configuration in which at least a portion of the thermal trigger is in direct contact with the thinned portion, to a second configuration due to current flowing through the conductive element being greater than a current threshold, wherein in response to the current flowing through the conductive element being greater than the current threshold, a temperature increase occurs at the thinned portion of the conductive element causing the deformation of the thermal trigger, the deformation of the thermal trigger from the first configuration to the second configuration operable to cause the conductive element to break at the thinned portion to stop current transfer therethrough, the thermal trigger being in direct contact with the location of the break of the thinned portion before breakage of the thinned portion.
2. The CID of claim 1, wherein the conductive element includes a fixed portion and a movable portion, the movable portion configured to move from a first position to a second position in response to deformation of the thermal trigger from the first configuration to the second configuration.
3. The CID of claim 2, wherein the conductive element includes a hinge disposed between the fixed portion and the movable portion, the hinge defining an axis of rotation about which the movable portion moves from the first position to the second position.
4. The CID of claim 1, wherein the thermal trigger includes at least one of a shape memory alloy

and a bi-metallic material.

5. The CID of claim 4, wherein the thermal trigger includes a nitinol-containing material.

6. A current-interrupt device (CID), comprising: a conductive element including a movable portion separated from a fixed portion by a hinge, and a thinned portion, the conductive element configured to be electrically coupled to a terminal of an electrochemical cell; and a thermal trigger positioned a distance from the conductive element in a first configuration, the thermal trigger dimensioned and configured to deform at a predetermined temperature from the first configuration in which at least a portion of the thermal trigger is in direct contact with the thinned portion, to a second configuration due to a current flowing through the conductive element being greater than a current threshold, wherein in response to the current flowing through the conductive element being greater than the current threshold, a temperature increase occurs at the thinned portion of the conducting element causing the deformation of the thermal trigger from the first configuration to the second configuration thereby causing the movable portion of the conductive element to move about the hinge such that conductive element breaks at the thinned portion to stop current transfer therethrough, the thermal trigger being in direct contact with the location of the break of the thinned portion before breakage of the thinned portion.

7. The CID of claim 6, wherein the thermal trigger includes at least one of a shape memory alloy and a bi-metallic material.

8. The CID of claim 7, wherein the thermal trigger includes a nitinol-containing material.

9. The CID of claim 6, wherein communication of an electrical current through the terminal is discontinued when the movable portion of the conductive element moves about the hinge.

10. A current interrupt device (CID) comprising: a housing; a conductive element coupled to the housing and configured to be electrically coupled to a terminal of an electrochemical cell, the conductive element having a first portion having a first thickness and a second portion having a second thickness less than the first thickness; and a thermal trigger configured to deform at a predetermined temperature from a first configuration in which at least a portion of the thermal trigger is in direct contact with the second portion, to a second configuration due to a current flow through the conductive element being greater than a current threshold, wherein in response to the current flowing through the conductive element being greater than the current threshold, a temperature increase occurs at the second portion of the conducting element causing the deformation of the thermal trigger, the deformation from the first configuration to the second configuration operable to cause the conductive element to break at the second portion to stop current transfer therethrough, the thermal trigger being in direct contact with the location of the break of the second portion before breakage of the second portion.

11. The CID of claim 10, wherein the first portion of the conductive element includes a fixed portion and a movable portion, the movable portion configured to move from a first position to a second position in response to deformation of the thermal trigger from the first configuration to the second configuration.

12. The CID of claim 11, wherein the conductive element includes a hinge disposed between the first portion and the second portion, the hinge defining an axis of rotation about which the movable portion moves from the first position to the second position.

13. The CID of claim 12, wherein the conductive element is disposed in an internal volume defined by the housing.

14. The CID of claim 13, wherein the housing defines a first aperture through which the conductive element is electrically coupled to the terminal of the electrochemical cell, and a second aperture located proximate to the movable portion such that the movable portion is disposed through the second aperture in the second configuration.

15. The CID of claim 12, wherein the thermal trigger is disposed within the internal volume.

16. The CID of claim 10, wherein the thermal trigger includes at least one of a shape memory alloy

or a bi-metallic material.

17. The CID of claim 16, wherein the thermal trigger includes a nitinol-containing material.
